TopoMoE: Decentralized Mixture-of-Experts Architecture via Tensor Topology Optimization and Information Flow Conservation

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Abstract

Mixture-of-Experts (MoE) architectures enable efficient scaling of large models via sparse activation, but traditional schemes suffer from centralized gating, lack of structural constraints on reasoning processes, and ambiguous credit assignment in weakly supervised settings. These limitations make them unsuitable for high-stakes domains such as healthcare and unmanned combat. This paper presents TopoMoE, a decentralized MoE architecture that integrates tensor topology optimization with topological information flow conservation.

Building upon the general *Topology-GRPO* framework proposed in the Universal Topological Agent theory, we introduce **KT-GRPO (Kernelized Tensor Topology-GRPO)**, a tensor-enhanced instantiation tailored for sequential reasoning in MoE systems. It lifts reasoning trajectories into a high-dimensional coupled tensor via Kronecker products, and uses Tucker low-rank compression to enable topology-aware policy optimization without scalar reward engineering.

At the multi-expert coordination level, we construct a discrete topological information flow field, using divergence conservation to ensure local logical consistency and curvature holonomy checks to enforce global closed-loop reasoning, eliminating centralized gating entirely. We formulate a unified differentiable objective that couples local policy gradients with global topological constraints, forming an end-to-end trainable paradigm of topological self-supervised learning.

This paper provides a complete theoretical framework, mathematical formulation, experimental protocol, and application analysis. The architecture achieves a paradigm shift from data-driven learning to structure-consistency-driven learning, serving as a next-generation decentralized foundation for complex decision-making, autonomous systems, and unmanned combat.

Keywords: Mixture-of-Experts; Tensor Topology; KT-GRPO; Topology-GRPO; Information Flow Conservation; Decentralized Autonomous Systems

1 Introduction

1.1 Background and Motivation

Mixture-of-Experts architectures have become a mainstream paradigm for efficient large-model scaling, balancing capacity and computation via dynamic sparse activation. However, conventional MoE suffers from critical drawbacks:

First, reasoning is largely black-box and optimized only for final outputs, which cannot distinguish valid logical reasoning from accidental prediction;

Second, global gating mechanisms introduce communication bottlenecks, poor robustness, and limited interpretability;

Third, in low-label, high-safety scenarios such as autonomous combat and healthcare, standard supervision and scalar-reward RL are inefficient and untrustworthy.

Recently, the *Universal Topological Agent* framework proposed *Topology-GRPO* as a general policy optimization paradigm that injects topological structure into reinforcement learning. However, this general formulation is not yet specialized for high-dimensional sequential reasoning and distributed autonomous collaboration scenarios. To fill this gap, we extend Topology-GRPO into a tensor-based form and embed it within a decentralized MoE backbone with geometric consistency constraints.

1.2 Core Contributions

1. Propose **KT-GRPO**, a **tensor-specialized instantiation** of the general Topology-GRPO algorithm, designed explicitly for sequential trajectory modeling in distributed autonomous systems;

2. Construct trajectory-outcome coupled tensors via Kronecker products to preserve intrinsic topological structure in decision-making chains;

3. Build a two-layer geometric consistency constraint: divergence conservation for local logic and curvature holonomy for global closed-loop consistency;

4. Develop a fully decentralized, end-to-end trainable topological MoE architecture with unified self-supervised learning;

5. Verify the architectural adaptability to unmanned combat systems from a theoretical perspective, expanding the application boundary of the framework.

2 Related Work

2.1 Mixture-of-Experts Architectures

Traditional MoE models such as GShard and Switch Transformer rely on centralized gating, leading to single-point failures and poor interpretability. Recent works such as TA-MoE and HMoE improve efficiency but ignore topological structure in collaborative reasoning. None integrate geometric consistency constraints for distributed anti-destruction collaboration.

2.2 Topology-GRPO and Policy Optimization

The Universal Topological Agent framework introduces **Topology-GRPO** as a general topology-aware reinforcement learning paradigm that stabilizes training via structural constraints. However, it operates on scalar or low-dimensional advantages and is not designed for distributed tensor-based reasoning. This work extends it to **KT-GRPO** for high-dimensional autonomous collaborative scenarios.

2.3 Geometric Topology in Deep Learning

Discrete differential geometry and graph-based regularization have been used to improve structure awareness, but few unify tensor-level trajectory modeling, decentralized MoE, and geometric conservation laws. TopoMoE closes this gap for safety-critical autonomous systems.

3 TopoMoE Architecture Design

3.1 Architecture Overview

TopoMoE models the collaborative intelligent system as a discrete topological manifold: functional nodes as experts, information interaction as directed information flow. The architecture consists of two core components:

(1) Single-expert decision optimization via **KT-GRPO**, a tensor-enhanced version of Topology-GRPO;

(2) Multi-expert distributed coordination via divergence and curvature constraints on the information flow field.

The entire system is trained end-to-end under a unified loss function, achieving local optimization and global consistency.

3.2 Single-Agent Optimization: KT-GRPO

3.2.1 From Topology-GRPO to KT-GRPO

The general **Topology-GRPO** algorithm provides a topological framework for stable policy optimization. To adapt it to sequential distributed decision-making, we propose its tensor-specialized form: **Kernelized Tensor Topology-**

GRPO (KT-GRPO), which encodes complete decision trajectories into high-dimensional tensor space to retain temporal and structural topological features.

3.2.2 Trajectory-Outcome Coupled Tensor

For a decision trajectory $\tau_i = (h_1^{(i)}, \dots, h_N^{(i)})$ and terminal decision embedding $\mathbf{y}^{(i)}$, we construct a coupled tensor via Kronecker product:

$$\mathcal{T}_i = \left(\bigotimes_{t=1}^{N} h_t^{(i)} \right) \otimes \mathbf{y}^{(i)}$$

This tensor explicitly preserves the topological correlation between sequential decision steps and final outputs, avoiding the loss of structural information in scalar reward optimization.

3.2.3 Tucker Low-Rank Compression

To reduce computational complexity, we perform Tucker low-rank decomposition on the coupled tensor:

$$\mathcal{T}_i \approx \mathcal{G}_i \times_1 U_1 \times_2 \cdots \times_{N+1} U_{N+1}$$

where $\mathcal{G}_i$ is the tensor core, and $U_1, U_2, \dots, U_{N+1}$ are factor matrices.

3.2.4 Tensor-Based Group Relative Advantage

Following the group-relative optimization logic of Topology-GRPO, we compute the relative advantage in tensor core space:

$$A_i^{\mathrm{KT}} = \frac{\|\mathcal{G}_i - \bar{\mathcal{G}}\|_F - \mu_{\mathcal{G}}}{\sigma_{\mathcal{G}}}$$

where $\bar{\mathcal{G}}$ is the mean tensor core, $\mu_{\mathcal{G}}$ and $\sigma_{\mathcal{G}}$ are the mean and standard deviation of tensor core norms.

3.2.5 KT-GRPO Policy Gradient

The policy gradient of KT-GRPO is defined as:

$$\nabla_\theta J_{\mathrm{KT}}(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[A^{\mathrm{KT}}(\tau) \cdot \nabla_\theta \log \pi_\theta(\tau) \right]$$

3.3 Multi-Agent Coordination: Topological Information Flow

3.3.1 Expert Potential and Information Flow

Define the potential of each expert node as ϕ_v, and the directed information flow between nodes is:

$$f_{vu} = \sigma_{vu} \cdot (\phi_v - \phi_u)$$

where σ_{vu} is the information transmission coefficient.

3.3.2 Discrete Divergence Conservation

The discrete divergence of the information flow field is:

$$(\nabla \cdot \mathbf{f})_v = \sum_u f_{vu} - \sum_u f_{uv}$$

The divergence conservation loss ensures local information balance:

$$\mathcal{L}_{\text{div}} = \mathbb{E}_{\mathbf{x}\sim\mathcal{D}} \left\| \nabla \cdot \mathbf{f}(\mathbf{x}) - \mathbf{s}(\mathbf{x}) \right\|_2^2$$

3.3.3 Discrete Curvature Holonomy Constraint

To ensure global closed-loop logical consistency, the curvature holonomy constraint of the loop path is constructed:

$$\Omega_{ijk} = \phi_i - \mathcal{P}_{ji}\phi_j - \mathcal{P}_{ki}\mathcal{P}_{kj}\phi_k$$

The corresponding loss function is:

$$\mathcal{L}_{\text{curv}} = \sum_{(i,j,k)\in\mathcal{F}} w_{ijk} \|\Omega_{ijk}\|_F^2$$

3.4 Unified Loss Function

The total loss function coupling local optimization and global topological constraints is:

$$\boxed{\begin{aligned} \mathcal{L}_{\text{total}}(\theta, \sigma) = &- \mathbb{E}_{\{\tau_i\}\sim\pi_\theta} \left[\sum_{i=1}^{G} A_i^{\text{KT}} \log \pi_\theta(\tau_i) \right] \\ &- \lambda\mathcal{L}_{\text{div}} - \mu\mathcal{L}_{\text{curv}} - \gamma\|\sigma\|_2^2 \end{aligned}}$$

3.5 Topological Self-Supervised Learning

TopoMoE takes topological structural consistency as the self-supervision signal, without relying on a large number of manual annotations, realizing stable policy learning in weak-supervision scenarios.

4 Experimental Protocol and Evaluation Design

4.1 Baseline Selection

1. Standard sparse MoE: GShard, Switch Transformer 2. Topology-aware RL: Graph-GRPO, general Topology-GRPO 3. Advanced MoE variants: HMoE, TA-MoE 4. Dense Transformers of equivalent scale

4.2 Evaluation Metrics

- General Metrics: Decision accuracy, sample efficiency, system throughput - Topological Consistency: Cross-node logical contradiction rate, divergence deviation, curvature holonomy deviation - Engineering Metrics: Convergence speed, computational overhead, communication volume

4.3 Implementation Details

Unified model backbone, parameter scale, training steps and optimizer are adopted for all baselines. The experimental environment is PyTorch + NVIDIA A100. Weakly supervised experiments are set with 10

4.4 Ablation Study

1. Full TopoMoE 2. TopoMoE w/o KT-GRPO (replaced with vanilla Topology-GRPO) 3. TopoMoE w/o curvature holonomy constraint 4. TopoMoE w/o topological information flow constraints

4.5 Visualization Scheme

1. Tensor trajectory clustering visualization via t-SNE / PCA 2. Information flow field, divergence heatmap and curvature distribution visualization
Note: This paper focuses on theoretical architecture and mathematical formulation, and all experimental protocols are for theoretical verification. No empirical test results or performance comparison data are given in this study.

5 Theoretical Application to Unmanned Combat Systems

5.1 Strategic Adaptability

The TopoMoE framework is highly compatible with the core requirements of modern **unmanned combat systems (UCS)**. Its decentralized topological design perfectly matches the anti-destruction, anti-interference and distributed collaboration needs of strong confrontation battlefields, solving the inherent defects of traditional centralized command architectures.

5.2 Decentralized Anti-Destruction Combat Command

Abandoning the centralized gating mechanism, each unmanned combat platform is modeled as an independent expert node, and the system realizes autonomous coordination through topological information flow consensus. Even if individual nodes are destroyed or communication is interrupted, the remaining nodes can still maintain stable operation, realizing anti-beheading and persistent combat capabilities.

5.3 Battlefield Information Consistency Assurance

The discrete divergence conservation and curvature holonomy constraints strictly restrict the transmission and interaction of battlefield perception information, which can effectively eliminate information conflicts, false target interference and data disorder, avoid fratricide caused by information distortion, and ensure the logical self-consistency of the entire combat system.

5.4 Auditable and Interpretable Autonomous Decision-Making

KT-GRPO encodes the complete decision-making trajectory of unmanned platforms into tensor space, realizing traceable and verifiable autonomous decision-making. It meets the audit and ethical constraints of lethal autonomous weapons systems, and solves the safety governance problem of black-box decision-making.

5.5 Weak-Supervision Autonomous Learning in Battlefield

Relying on topological self-supervised learning, the system can realize autonomous policy optimization without a large number of labeled battlefield data, adapting to the characteristics of weak communication, less annotation and strong real-time performance in deep penetration and silent combat missions.

5.6 Heterogeneous Unmanned Swarm Collaboration

The framework can unify heterogeneous combat units such as unmanned aerial vehicles, loitering missiles, electronic warfare platforms and ground unmanned vehicles into the same topological manifold, realizing cross-domain and multi-modal intelligent collaboration of combat resources, and supporting the integrated operations of air-ground-electromagnetic domains.

6 Analysis and Discussion

6.1 Topological Consistency Mechanism

KT-GRPO inherits the topological stability of general Topology-GRPO, and the tensorized trajectory modeling enhances the structural perception of sequential decisions. The two-layer information flow constraint realizes the unification of local optimization and global consistency, which is the core mechanism for the framework to adapt to safety-critical scenarios.

6.2 Advantages of Decentralized Architecture

The decentralized design eliminates single-point failures, reduces communication bottlenecks, and improves the robustness and scalability of the system. The topological constraint mechanism makes the system behavior interpretable and controllable, which is more suitable for high-safety autonomous collaboration scenarios.

6.3 Limitations and Future Work

- Curvature holonomy calculation brings certain computational overhead in large-scale node networks; - Tensor rank and constraint coefficients need manual tuning; - Future work will focus on adaptive topology optimization, lightweight constraint calculation and domain-specific topological prior embedding.

7 Conclusion

This paper proposes TopoMoE, a decentralized MoE architecture built on topological principles. By introducing **KT-GRPO**—a tensor-specialized instantiation of the general Topology-GRPO framework—and enforcing geometric conservation laws, TopoMoE achieves structured, consistent, and self-supervised learning. The framework has inherent theoretical adaptability to unmanned combat systems, and provides a reliable, interpretable and anti-destructive theoretical foundation for distributed autonomous collaborative systems.

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